



B.M.S. College of Engineering, Bengaluru-560019

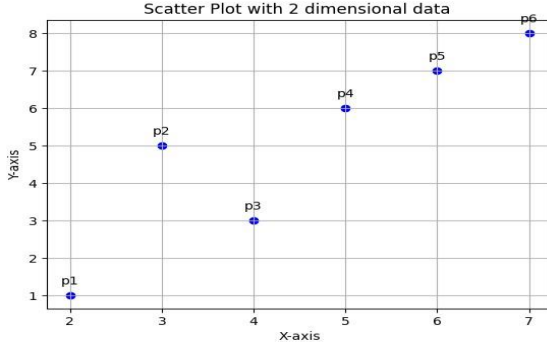
(Autonomous Institute Affiliated to VTU)

DEPARTMENT OF MATHEMATICS

Course Title: Linear Algebra and optimization Course Code: 23MA4BSLAO CIE: III SEM: IV-CSE stream(Except AIML)		
Question Number	Solution	Marks Allocated
1	Verify that set $S = \{(x_1, x_2)/3x_1^2 + 2x_2^2 \leq 6\}$ is Convex or not. Soln: Let x, y be two points of S where $x = (x_1, x_2)$ and $y = (y_1, y_2)$ The line segment joining x and y is the set $\{u: u = \lambda x + (1 - \lambda)y: 0 \leq \lambda \leq 1\}$ For some λ, let $u = (u_1, u_2)$ be the point of this set So $u_1 = \lambda x_1 + (1 - \lambda)y_1, u_2 = \lambda x_2 + (1 - \lambda)y_2$ Now $3u_1^2 + 2u_2^2 = 3(\lambda x_1 + (1 - \lambda)y_1)^2 + 2(\lambda x_2 + (1 - \lambda)y_2)^2$ $\leq 6\lambda^2 + 6(1 - \lambda)^2 + 12\lambda(1 - \lambda)$ ≤ 6 Thus $S = \{(x_1, x_2)/3x_1^2 + 2x_2^2 \leq 6\}$ is convex set.	2 1 3
2	Determine $\nabla \begin{bmatrix} xe^{2y} \cos(z) \\ (y - 2)^2 \\ \sin\left(\frac{z}{2}\right), e^{2y} \cos(z) \end{bmatrix}$ w.r. to (x, y, z) at $(3, 0, \pi)$. Soln: $f = xe^{2y} \cos(z), g = (y - 2)^2 \sin\left(\frac{z}{2}\right), h = e^{2y} \cos(z)$ First order partial derivatives are $f_x = e^{2y} \cos(z), f_y = 2xe^{2y} \cos(z), f_z = -xe^{2y} \sin(z)$ $g_x = 0, g_y = (2y - 4) \sin\left(\frac{z}{2}\right), g_z = \frac{(y - 2)^2 \cos\left(\frac{z}{2}\right)}{2}$ $h_x = 0, h_y = 2e^{2y} \cos(z), h_z = -e^{2y} \sin(z)$ At the point $(3, 0, \pi)$. The required matrix is $\begin{bmatrix} -0.99 & -5.99 & 0 \\ 0 & -3.99 & 0 \\ 0 & -1.99 & 0 \end{bmatrix}$	4 2
3	Find the singular value decomposition of the matrix $A = \begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & -2 \end{bmatrix}$. Soln: $A^T A = \begin{bmatrix} 9 & -9 \\ -9 & 9 \end{bmatrix} = B$ $ B - \lambda I = 0$ $\Rightarrow \lambda^2 - 18\lambda = 0$ $\lambda = 18, 0$ when $\lambda = 18 \Rightarrow X = [1, -1]^T$	2

	when $\lambda = 0 \Rightarrow X = [1, 1]^T$ $\Rightarrow \epsilon = \begin{bmatrix} \sqrt{18} & 0 \\ 0 & 0 \end{bmatrix}, V^T = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$ $U = \begin{bmatrix} \frac{1}{3} & 0 & 2/\sqrt{5} \\ -2/3 & 1/\sqrt{2} & 1/\sqrt{5} \\ 2/3 & 1/\sqrt{2} & 0 \end{bmatrix}$ $A = U \in V^T$	3 2
4	Find the optimum value of the function $f(x, y, z) = -9x^2 + 6xy - 2y^2 - 2xz - 2z^2$ by verifying the nature of critical points using hessian matrix. Soln: $f_x = -18x + 6y - 2z = 0, f_y = 6x - 4y = 0, f_z = -2x - 4z = 0$ $\Rightarrow P(0, 0, 0)$ $f_{xx} = -18, f_{xy} = 6, f_{xz} = -2$ $f_{yx} = 6, f_{yy} = -4, f_{yz} = 0$ $f_{zx} = -2, f_{zy} = 0, f_{zz} = -4$ The required Hessian matrix $H = \begin{bmatrix} -18 & 6 & -2 \\ 6 & -4 & 0 \\ -2 & 0 & -4 \end{bmatrix}$ The eigen values are $\lambda = -4, -\sqrt{89} - 11, \sqrt{89} - 11 < 0$ $\Rightarrow$ critical points are contains maximum value $\Rightarrow f(0, 0, 0) = 0$	2 2 2 1
5	For the given $f = \begin{bmatrix} e^{x_0^2 x_1 x_2} & \ln(x_1^2 x_2 x_3) \\ x_2^3 x_1 x_3 & x_2^3 x_1 \end{bmatrix}$ and $X = \begin{bmatrix} x_0 & x_1 \\ x_2 & x_3 \end{bmatrix}$ verify $\frac{\partial}{\partial X}(\text{Trace}(f(X))) = \text{Trace}\left(\frac{\partial f(X)}{\partial X}\right).$ Soln: $\text{Trace}(f(X)) = e^{x_0^2 x_1 x_2} + x_2^3 x_1$ $\frac{\partial}{\partial X}(\text{Trace}(f(X))) = \begin{bmatrix} e^{x_0^2 x_1 x_2} \cdot 2x_0 x_1 x_2 \\ e^{x_0^2 x_1 x_2} \cdot x_0^2 x_2 + x_2^3 \\ e^{x_0^2 x_1 x_2} \cdot x_0^2 x_1 + 3x_1 x_2^2 \\ 0 \end{bmatrix} \rightarrow LHS$ $\frac{\partial f(X)}{\partial X} = \begin{bmatrix} \begin{bmatrix} e^{x_0^2 x_1 x_2} \cdot 2x_0 x_1 x_2 & 0 \\ 0 & 0 \end{bmatrix} & \begin{bmatrix} e^{x_0^2 x_1 x_2} \cdot x_0^2 x_1 & \frac{1}{x_2} \\ 3x_2^2 x_1 x_3 & 3x_2^2 x_1 \end{bmatrix} \\ \begin{bmatrix} e^{x_0^2 x_1 x_2} \cdot x_0^2 x_2 & \frac{2}{x_1} \\ x_2^3 x_3 & x_2^3 \end{bmatrix} & \begin{bmatrix} 0 & \frac{1}{x_3} \\ x_2^3 x_1 & 0 \end{bmatrix} \end{bmatrix}$ $\text{Trace}\left(\frac{\partial f(X)}{\partial X}\right) = \begin{bmatrix} e^{x_0^2 x_1 x_2} \cdot 2x_0 x_1 x_2 \\ e^{x_0^2 x_1 x_2} \cdot x_0^2 x_2 + x_2^3 \\ e^{x_0^2 x_1 x_2} \cdot x_0^2 x_1 + 3x_1 x_2^2 \\ 0 \end{bmatrix} \rightarrow RHS$ OR	1 2 4

	$\frac{\partial f(X)}{\partial X} = \begin{bmatrix} \begin{bmatrix} e^{x_0^2 x_1 x_2} \cdot 2x_0 x_1 x_2 & e^{x_0^2 x_1 x_2} \cdot x_0^2 x_2 \\ e^{x_0^2 x_1 x_2} \cdot x_0^2 x_2 & 0 \end{bmatrix} & \begin{bmatrix} 0 & \frac{2x_1 x_2 x_3}{x_1^2 x_2 x_3} \\ \frac{x_1^2 x_3}{x_1^2 x_2 x_3} & \frac{x_1^2 x_2}{x_1^2 x_2 x_3} \end{bmatrix} \\ \begin{bmatrix} 0 & x_1^2 x_3 \\ 3x_2^2 x_1 x_3 & x_2^3 x_1 \end{bmatrix} & \begin{bmatrix} 0 & x_2^3 \\ 3x_2^2 x_1 & 0 \end{bmatrix} \end{bmatrix}$ $Trace\left(\frac{\partial f(X)}{\partial X}\right) = \begin{bmatrix} e^{x_0^2 x_1 x_2} \cdot 2x_0 x_1 x_2 & e^{x_0^2 x_1 x_2} \cdot x_0^2 x_2 + x_2^3 \\ e^{x_0^2 x_1 x_2} \cdot x_0^2 x_1 + 3x_1 x_2^2 & 0 \end{bmatrix} \rightarrow \text{RHS}$ $Trace(f(X)) = e^{x_0^2 x_1 x_2} + x_2^3 x_1$ $\frac{\partial}{\partial X}(Trace(f(X))) = \begin{bmatrix} e^{x_0^2 x_1 x_2} \cdot 2x_0 x_1 x_2 & e^{x_0^2 x_1 x_2} \cdot x_0^2 x_2 + x_2^3 \\ e^{x_0^2 x_1 x_2} \cdot x_0^2 x_1 + 3x_1 x_2^2 & 0 \end{bmatrix} \rightarrow \text{LHS}$	4 1 2
6 a	Let H is the hyperplane that passes through the points of the set $A = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 5 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}$ and G be the hyperplane that passes through the points of the set $B = \left\{ \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right\}$ then find the angle between two hyperplanes H and G. Soln: $A = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 5 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}$ $n \cdot x = d \Rightarrow ax + by + cz = d$ then $a + c = d, 2a + b + 5c = d, a + b + c = d$ $[A:B] = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 2 & 1 & 5 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$ By reducing into echelon form $\begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 3 & -1 \\ 0 & 0 & -3 & 1 \end{bmatrix}$ $\Rightarrow a = \frac{4}{3}, b = 0, c = 1$ Then the equation of the plane H is $4x - z = 3 \dots (1)$ $B = \left\{ \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right\}$ $-a + 2c = d, b + c = d, a + b = d$ $[A:B] = \begin{bmatrix} -1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{bmatrix}$ By reducing into echelon form $\begin{bmatrix} -1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$ $\Rightarrow a = 1, b = 0, c = 1$ Then the equation of the plane G is $x + z = 1 \dots (2)$ The angle between the plane	3 3 1

	$\cos(\theta) = \frac{n_1 \cdot n_2}{ n_1 \cdot n_2 } \Rightarrow \theta \approx 59^\circ$ Where n_1 & n_2 is normal to the hyper planes and G respectively	
6b	Apply Principal Component Analysis to project the given 2-dimensional data on 1-dimension.  Soln: Covariance matrix $S = \begin{bmatrix} 1 & 0.9019 \\ 0.9019 & 1 \end{bmatrix}$ Eigen values are $\lambda = 1.9019, 0.09809$ Eigen vectors are for $\lambda_1 = 1.9019$ are $[0.7071, 0.7071]$ The corresponding 1-D data are: $[-1.3363, -8018, -0.2673, 0.2673, 0.8018, 1.3363]$	2 1 2 2

NOTE: Appropriate marks to be awarded for correct alternate methods.

Scheme and solutions prepared by:

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